

Time after time: neural codes of speech comprehension

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Abstract

The acoustic signal of speech is fleeting and unfolds linearly; yet, listeners must derive temporally extended, hierarchically organised, linguistic structures to comprehend it. This article reviews evidence that the human brain transforms the transient auditory signal into neural representations that are persistent, parallel, and time-stamped. First, neural persistence allows information to remain available after it has disappeared from the acoustic input. Second, parallel encoding of recent inputs, across multiple levels of language structure, enables composition and interactions across the speech hierarchy. Third, dynamic neural representations encode both content and elapsed time, providing a flexible mechanism for preserving sequence order. Together, these representational properties reveal how the human brain transforms a continuously disappearing signal into temporally extended linguistic structure.

Keywords: *neuroscience; speech; perception; dynamics; encoding; sequencing*

The transient life of speech

If I were to read this sentence aloud, you would hear a stream of sound that is spectrally complex and fluctuates rapidly across a broad range of frequencies over its few-second duration. As speech unfolds, each new word succeeds the previous one in the acoustic signal, and once the utterance ends, the signal is gone. This means that the listener cannot replay earlier parts of an utterance, unless they ask the speaker to repeat them. This differs fundamentally from written language, which allows readers to linger on words, reread, and revisit inputs in any order; in speech, the input is transient, and its sequence is fixed by the flow of time rather than the will of the comprehender.

This transience poses a significant challenge: constructing stable representations from a signal that is continuously disappearing. For example, speech comprises many levels of **hierarchical structure** (see Glossary), ranging from lower-level units such as phonemes and syllables, to mid-level linguistic units relating to individual morphemes and words, to high-level units such as phrases and sentences (see Box 1). These levels unfold over different timescales, whereby lower-level units are shorter and unfold more quickly, and higher-level units extend much longer over time.

Successful speech **comprehension** therefore depends upon the brain's ability to assemble hierarchical linguistic structure, expressed within nested timescales, from a transient and short-lived auditory signal. This article reviews work demonstrating that the brain's solution to this challenge depends on three fundamental properties of neural representations: persistent, parallel, and time-stamped encoding. By constructing representations that preserve information over time, coexist in parallel, and encode temporal relationships between elements, the brain can overcome the limitations imposed by the transient nature of speech.

Persistence over time

The idea of **persistent encoding** is simple. Rather than the brain only processing information while the inputs are physically exerting influence on the sensory system, the brain continues to process and encode those inputs past the point at which they have dissipated from the acoustic signal (see **Figure 1**). This turns the externally transient signal into an internally enduring one.

It is well established that neural responses outlast very simple auditory inputs such as brief clicks and tones. Seminal studies report that even when the auditory stimulus lasts just 5-50 ms, cortical responses last hundreds of milliseconds [1]. Furthermore, sensory memory for these simple auditory stimuli endures for several seconds, as measured through discrimination behaviour and auditory ERPs to deviant tones [2].

Recent studies build on this foundational work by demonstrating that persistence occurs beyond responses to simple sounds. During continuous speech, information about complex spectro-

temporal inputs, as well as more abstract information across the **speech hierarchy** (e.g., was it a /p/ or a /b/; a noun or a verb) remains encoded for extended periods of time.

The average phoneme in English lasts only 80-100 ms [3]; however, evidence from electro- (EEG) and magneto-encephalography (MEG) shows that the phonetic features of these sounds are neurally encoded for approximately 400 ms [4–6]. Persistence of speech encoding is also present at the lexical level, where the average word in English lasts approximately 350 ms [7,8]. For example, in a study that examined a suite of lexical representations during continuous story listening, the authors found that higher-order properties of speech, such as semantic and syntactic features, could be decoded from neural activity for 1-2 seconds after the word had ended [9]. Other studies have also shown persistent encoding of word frequency and grammatical class [9–13], as well as word meaning and probabilities over candidate words [14], which are encoded for a second or longer [9,10,15–19].

As can be appreciated when comparing the timescales of phoneme and word processing, the duration of persistent encoding is not uniform across the speech hierarchy. Instead, it scales with representational level: lower levels of the hierarchy are processed for a shorter period of time, and higher levels are processed for longer periods [20,21]. This hierarchical scaling of persistent encoding duration could not be established from earlier studies focusing solely on low-level auditory processing.

Prolonged access to speech input alleviates several of the pressures imposed by a transient signal. It allows the brain to carry information forward, both to anticipate what is about to happen [22–24], and to update interpretations of what happened in the past based on what happens later [25–28]. This latter idea – where *subsequent* phonemes and words serve to update *previous* phonemes and words – is referred to as “retrodictive” or **postdictive processing** [6,29–31]. It is an interesting phenomenon that sounds almost sci-fi in its description – information in the future is used to re-interpret the past, as if the listener were travelling back in time to revise perception. The reality, of course, is that the listener does not travel to the past, but rather that the preceding information is maintained or carried into the future so that it can be disambiguated later. Once a listener hears the disambiguating information, they automatically perceive the correct earlier phoneme in the sequence, without being aware that any adjustment to perception has occurred [24,32].

At the phoneme level, researchers have developed clever experimental paradigms to study postdictive processing. In these experiments, spoken words contain an ambiguous phoneme at word onset (for example, a phoneme midway between /p/ and /b/), and the phoneme’s interpretation can only be disambiguated later, once the identity of the word is revealed (e.g., “parakeet”). These stimuli are known as “lexical garden paths”. Studies have found that listeners encode information about the initial phoneme while the rest of the word is being heard and processed [6,30]. Recent work sharpens this account by suggesting that neural activity in superior temporal gyrus (STG) maintains multiple copies of the initial phoneme: the original acoustic evidence and an updated interpretation informed by the later context [24,33,34,35]. Competing interpretations can therefore remain available until ambiguity is resolved [36,37].

In addition to informing the dynamics of information encoding during speech processing, postdictive studies provide critical insight into the **representational format** of the information that is encoded over time. The brain appears to preserve fine-grained acoustic evidence while deriving more abstract phoneme representations: persistence is not simply maintenance of the original veridical input but continued representational transformation and derivation. Building a full understanding of the different representational formats that are derived, however, is often challenging because properties at different representational levels are highly correlated. As a result, a decoder trained to discriminate phonetic features will likely recover them even if the underlying neural representation shifts from an acoustic code to an abstract categorical code. Persistent decodability does not therefore, by itself, establish which details of the original speech input remain represented, or whether the underlying representational format is changing over time.

Persistence in service of ambiguity resolution occurs at several levels of linguistic description – not just at the level of phonemes. The interpretation of a word, for instance, also depends on its relation to words that occur later in the sentence. This is true both in terms of interpreting a word's meaning (its semantics) and interpreting its role in the sentence structure (its syntax). For example, the word *bat* remains ambiguous in the phrase "*She looked at the bat and watched it disappear into the cave*" until later context reveals whether it refers to the animal or sports equipment. Neural studies of lexical ambiguity have shown that ambiguous words initially activate multiple candidate meanings, with subsequent context biasing interpretation towards the meaning most consistent with the sentence as a whole [22,38,39]. At the syntactic level, in sentences such as "*While Anna dressed the baby played in the crib*", it is only revealed later in the sentence structure that the baby is the subject of a new clause rather than the object of *dressed*. Such reanalysis is reflected in P600 responses and activity within inferior frontal and temporal cortices [40–44]. Interestingly, even after encountering disambiguating information, listeners exhibit continued sensitivity to initial or alternative interpretations of the sentence, even endorsing interpretations that are consistent with the initial misanalysis rather than the ultimately correct sentence meaning [45]. This suggests that alternative interpretations can remain active in parallel, allowing later information to revise earlier analyses without necessarily eliminating them entirely.

Together, these findings indicate that the language system maintains information about phonemes and words long after they have dissipated from the sensory input. Although speech unfolds as a transient acoustic signal, its neural representations persist over extended periods of time. Persistent representations allow the system to continuously update, refine, and compose linguistic structure as speech unfolds.

Parallelism over inputs and levels

Parallel encoding, the ability to simultaneously process multiple inputs across multiple levels of abstraction, is a bedrock of the speech comprehension system [47,48]. If neural responses outlast the acoustic events that elicited them, then subsequent inputs are, by definition, being processed while earlier ones remain represented in the system. However, as will be discussed, coexistence

alone is not enough for a successful parallel process - the system must support **recoverable representations** that remain available for downstream computation and do not interfere with one another. Persistent encoding is therefore necessary, but not sufficient, for parallel processing.

There are two axes over which the brain parallelises, which serve different computational functions. First is parallel processing of inputs in a sequence, which enables **composition** into higher order structures. Second is parallel processing across the speech hierarchy, which enables interactions between levels of representation to derive the appropriate interpretation of the input.

Parallel encoding of inputs

At any given moment, the brain is not just encoding the input that is being heard “right now”, but also a recent history of preceding inputs. For example, upon hearing the final sound /g/ in the word “dog”, the brain is simultaneously encoding the local sequence of phonemes – /d/, /o/, and /g/.

Neurophysiological evidence suggests that this phenomenon extends across multiple levels of the linguistic hierarchy. In an MEG study of audiobook listening, decoding analyses revealed that the brain simultaneously encodes the previous three-to-four phonemes of the unfolding speech stream [5]. Similar findings have been reported at the lexical level, where the properties of approximately four preceding words could be decoded from a single snapshot of neural activity, indicating that multiple words remain concurrently represented over time [9,12]. These findings further extend earlier evidence of persistent responses to simple auditory stimuli by demonstrating a key consequence for speech comprehension – persistent encoding maintains a concurrently available history of recent linguistic inputs for ongoing processing, rather than just the current input in isolation. This temporally extended representational state contains information about multiple neighbouring elements simultaneously, and is continuously updating as new elements come in. This enables computations that operate over a local sequence, akin to an evolving N-gram representation [49–51].

Recent neurophysiological studies suggest that higher-order linguistic structure is assembled from these concurrently represented lower-level elements. For example, one study compared neural responses to spoken phrases (“de rode vaas” (the red vase)) and sentences (“de vaas is rood” (the vase is red)) [52]. Despite acoustic, phonemic, and lexical similarities, the authors observed distinct neural response dynamics to the phrase and sentence stimuli, consistent with the generation of higher order structure. Recent studies examining constituent building and dependency processing have also shown that unresolved linguistic relationships are represented until the information required to complete them becomes available [53–57]. Together, these findings offer new insight into how progressively larger linguistic structures are built from sequential elements through parallel encoding.

A remaining challenge is how multiple speech representations coexist without interference. One candidate mechanism is **rotational dynamics** - information is passed between different **neural dimensions** over time, separating maintained representations from new inputs. In mouse

auditory cortex, population activity evolves over time to “move” the neural representation into different orthogonal dimensions [58]. Recent human MEG studies extend this principle to natural speech - several recent phonemes and words can be recovered from a snapshot of neural activity, yet each occupies distinct rather than mixed activity patterns [5,9]. While these are new and active areas of discovery, such findings suggest that the key to parallel encoding of recoverable representations in speech comprehension may be the use of a dynamic spatio-temporal neural code, as opposed to a static and persistent one.

In sum, the brain processes sequential elements of the speech input in parallel. This is analogous to a reader seeing all the letters of a word simultaneously, rather than one letter at a time. By maintaining a local sequence of recent inputs within a dynamic code, the brain transforms a transient auditory signal into a format that supports composition, allowing elements occurring at different moments in time to be combined into words, phrases, and sentences.

Parallel encoding of levels

Speech comprehension requires more than maintaining multiple inputs within a sequence. The system must also coordinate information across levels of linguistic representation, from speech sounds to meaning.

Speech processing is often conceived as a progression through a hierarchy of increasingly abstract representations [37,59]. In this view, acoustic information is transformed into phonemes, phonemes into words, and words into larger linguistic structures. Many classic views posit that each stage of abstraction replaces the one before it [60]. However, recent evidence suggests that rather than discarding lower-level representations once higher-level representations have been derived, the brain maintains multiple levels of linguistic structure simultaneously [9,21,61]. At any given moment during comprehension, neural activity encodes acoustic, phonetic, lexical, and higher-order linguistic information at the same time [9–12,21]. For example, when hearing the sentence “*The fat cat disappeared*”, the final phoneme /t/ in “cat” is represented simultaneously as a speech sound, as part of a particular syllable and word, as part of a noun, and as a concept associated with a furry animal. These representations differ drastically in their level of abstraction, yet all are available at the same moment of processing (see **Figure 2**).

The simultaneous encoding of information at multiple linguistic levels allows those levels to interact. Higher-level knowledge can influence the interpretation of lower-level inputs, while lower-level sensory evidence can continue to influence higher-level interpretations. For example, prior knowledge and sentence context improve the intelligibility of degraded speech and alter neural responses within the auditory cortex [11,62,63]; sentence-level context can influence the interpretation of upcoming words before perceptual processing has concluded [12]; and discourse-level context can scaffold semantic interpretations of upcoming words, such as supporting an animate interpretation of a typically inanimate object [64]. These bidirectional interactions suggest that speech comprehension relies on a layered representational state in which multiple levels remain concurrently available, permitting information to be exchanged as meaning is constructed over time.

Taken together, these findings suggest that the brain encodes multiple inputs and multiple levels of linguistic structure simultaneously. This enables the integration of information across time and across levels of representation, supporting robust interpretation of a rapidly unfolding signal.

Time-stamps for ordered sequencing

The meaning of words and sentences depends not just on the content of individual elements, but also their order. The word “*pest*” means something different from “*pets*”; the sentence “*the dog ate the ice cream*” is different from “*the ice cream ate the dog*”. While the individual elements are the same, their order leads to fundamentally different interpretations and meanings. Therefore, to comprehend speech successfully, the system must preserve information not just about “what” inputs occurred, but also about “when” each input occurred relative to the others.

The challenge of keeping track of the relative position of elements in a sequence is not unique to speech comprehension – it is found across a range of cognitive and sensory domains, including working memory [65,66], episodic memory [67,68], navigation [69,70], motor sequencing [71,72], among others. In animal models, neural populations have been shown to implement a **time code** for elapsed processing time (see **Box 2**). For example, hippocampal population activity in rodents contains neurons whose firing reflects elapsed time since an event occurred, providing a mechanism for linking elements within a sequence according to their temporal position [73–75]. Similar temporal coding schemes have been observed in other species and tasks, including song production in birds, where neurons in the premotor nucleus HVC fire at precise moments during learned vocal sequences to track progression through the song [76,77].

In the previous section, rotational dynamics were introduced as a mechanism for separating overlapping representations and protecting them from interference. The same dynamics may also solve the sequencing problem by time-stamping those representations [5,9,78,79]. As a representation rotates between neural dimensions, it traces a **neural trajectory** through **state space**, such that its location on the trajectory changes systematically with elapsed processing time. The current neural state therefore indicates both *what* information is represented and *when* the corresponding input occurred, thus time-stamping its content.

Recent studies provide direct population-level evidence for this second function of rotational dynamics during natural speech. Intracranial recordings of human superior temporal gyrus reveal that population activity tracks relative elapsed time within words, resulting in a neural trajectory that evolves continuously from word onset to offset [78]. This finding sharpens earlier non-invasive evidence that linguistic representations jointly encode content and elapsed time [9] by revealing that this trajectory is encoded by local cortical populations.

Together, these findings suggest that rotational dynamics serve two computational functions - separating overlapping representations and time-stamping those representations. Such a code provides a natural solution to the sequencing problem. If successive elements are associated with different temporal states, then their order can be recovered from the temporal relationships between those states (see **Figure 3**).

Importantly, this proposal differs from theories in which sequence position is represented explicitly through ordinal “slots” [80–82]. In these accounts, neural populations are selectively tuned to particular positions in a sequence, becoming active whenever an element occupies the preferred position, regardless of its timing. By contrast, recent evidence supports the idea that neural representations are organised primarily according to continuous unfolding time. Position can be derived from the temporal code, but ordinal position itself is not the native representational format [79].

Interestingly, similar computational solutions to encoding element position have emerged in artificial systems for processing language. The transformer architecture of large language models, for example, implements sinusoidal positional embeddings, in which the encoding of each token (roughly corresponding to a word) in the sequence is modulated by its phase in the sinusoid [83]. Similarly, rotary positional embeddings encode relative position through rotations of the input tokens as the sequence unfolds, allowing recovery of token position without depending on explicit position codes [84,85]. The neural code that represents linguistic information may undergo a similar rotation, and these computational methods offer a novel toolkit to probe and better understand the underlying geometric transformations occurring in the brain [86]. Overall, this reveals an intriguing convergence between the human brain and computational language processing systems - coding relative time alongside element content, which allows extraction of the position of elements in a sequence without depending on dedicated position-selective units.

In sum, internal representations are time-stamped, yielding a joint code for content and elapsed time since the input occurred. By embedding temporal context within evolving neural population dynamics, the brain can maintain access to recent inputs while preserving their order within the unfolding sequence. Sequence position then emerges from the temporal relationships between representations, rather than requiring a dedicated positional code.

Concluding Remarks

Speech comprehension depends on three fundamental properties of neural representations: persistence, parallelism, and time-stamped encoding. If we take this framework seriously, where should the field of speech neuroscience place its efforts to maximize discovery over the next decade?

Persistence and parallelism highlight that the human brain is incredibly reluctant to discard information. Traditional descriptions of language processing often imply a progression through

increasingly abstract representations, with each stage replacing the one before it. The present review highlights recent evidence against that position: previous inputs are encoded after subsequent ones arrive; lower-level representations remain available after higher-level representations are derived; and alternative interpretations coexist rather than being immediately removed upon resolution. From the brain's perspective, efficiency is not about representing as little information as possible, but rather preserving as much information as possible for future computation.

This representational greediness motivates a different set of questions for future research. Speech neuroscience has devoted substantial effort to identifying the "correct" representational primitives underlying language processing. The evidence reviewed here suggests that the brain is very willing to maintain multiple descriptions of the same input simultaneously, rather than committing to a single representational format. Rather than asking which representation wins, we may learn more by asking which representations coexist and how they interact.

Time-stamped encoding through state-space trajectories has been implicated in working memory, navigation, motor sequencing, and episodic memory across species. If the role of these trajectories is to preserve information about both content and time, then the sequencing problem faced by speech comprehension may be one instance of a broader computational challenge shared across species and domains. Historically, it was difficult to test this possibility directly because neural measurements in humans and animal models were so different. Using the same technologies across species, such as microelectrodes and Neuropixels arrays [87–89], opens the possibility to test whether the trajectories observed during speech comprehension reflect the same computational principle identified in other behaviours and organisms. This would allow the field to move beyond drawing algorithmic analogies between language and other domains, and instead directly test whether related computations are implemented by the same neural mechanisms.

Along those lines, single-cell electrophysiology has traditionally focused on understanding what information is encoded by individual neurons. However, if speech representations are encoded as trajectories through a neural state space, future insights will come not from cataloguing the tuning profiles of individual neurons, but from asking how single neurons participate in the broader population ensemble.

In addition to developing a better understanding of single-cell dynamics, a substantial challenge will be to reconcile these findings with meso- and macro-scale recordings, particularly the non-invasive measurements that constitute the majority of human neuroscience. Speech comprehension is implemented across spatial scales, from single neurons, local populations, cortical regions, and whole-brain networks; a comprehensive account of speech processing will therefore require connecting findings across these complementary scales. Dynamics at one scale do not uniquely determine their implementation at another. For instance, a representation that appears sustained in macro-scale signals such as EEG or MEG could arise from evolving neural generators at the level of individual neurons or local populations. Conversely, dynamics observed in single-neuron activity or local field potentials may not be detectable at the meso- or macro-

scales. Connecting these levels will be essential for moving toward a more complete account of joint spatio-temporal dynamics that considers multiple spatial scales simultaneously.

An important challenge for future work will concern the transition from linear sequences to hierarchical structure. Persistent and time-stamped encoding provides access to multiple neighbouring elements in an ordered linear sequence; however, words connect with each other hierarchically as well as linearly, and many linguistic dependencies are defined over abstract structural relationships rather than temporal adjacency. A central question is therefore how temporally extended representations are transformed into, or themselves support, hierarchical composition.

Studies are beginning to reveal how linguistic structure, statistical regularity, and variation in the dynamics of the speech input shape the neural dynamics of information processing (see Box 3). For example, when upcoming information is predictable, speech is processed slightly earlier [90]; when upcoming information is uncertain, it is processed slightly longer [5]. Characterising how neural dynamics change under different conditions may reveal what information the brain uses to initiate and guide processing. Examining these dynamics as listening conditions are pushed to their limits, and comprehension begins to break down, may further reveal the processing constraints that govern successful comprehension and, ultimately, the operating principles of the system as a whole [91].

Speaking more broadly, the framework outlined here primarily describes the neural *representations* the brain generates to support speech comprehension. The emerging picture is of a system that preserves information across time, levels of abstraction, and interpretation, creating a rich representational substrate from which linguistic structure can be composed. While representations are an important part of understanding the speech processing system, we lack an explanation of *how* those representations are neurally generated, and what algorithm is in place to orchestrate and compose those representations [92]. The next frontier will be to identify the computational circuit that operates over persistent, parallel, and time-stamped representations to ultimately derive complex linguistic structure and robust comprehension.

Figure 1. Persistent neural encoding. Speech unfolds as a transient acoustic signal in which each element is available only briefly. However, neural representations persist beyond the duration of the acoustic events that generated them. As successive words are heard, neural encoding of previous elements persists, creating a temporally extended neural record of recent speech input. This persistence allows later-arriving information to interact with and revise the interpretation of earlier inputs, providing opportunity for ambiguity resolution and the integration of information over time.

Figure 2. Parallel processing across linguistic levels. Speech comprises different levels of structure, from sounds up to meaning. As time unfolds, the brain represents information at multiple levels of abstraction simultaneously, coexisting in parallel rather than replacing one another. This layered representational state allows interactions between levels of the hierarchy, enabling higher-level knowledge to shape lower-level perception and lower-level sensory evidence to refine higher-level interpretations. Neural representations at different levels also persist for different durations, creating overlapping windows of information across the speech hierarchy.

Figure 3. Time-stamps for ordered sequencing. Linguistic representations evolve through neural state space as time unfolds (the coloured arrows), creating a joint code for content (the phonemes / words in the boxes) and elapsed time (the position of those boxes on the arrow trajectory). Sequence position emerges from the temporal relationships between population states. When queried at the end of an utterance, earlier elements are found further along the trajectory, and more recent ones are found towards the beginning. Similar coding principles can be applied across levels of the speech hierarchy, from phoneme sequences to word sequences, allowing the brain to preserve the order of elements within an unfolding linguistic structure.

Box 1: Speech hierarchy

Language is organised by a hierarchy of representational levels, which span the acoustic speech signal, phonemes, syllables, words, phrases, sentences, and discourse. Lower levels are less abstract and the units are shorter; higher levels are more abstract and the units are longer.

The lowest level is the acoustic signal, which can be described as a time-frequency representation akin to the information encoded at the cochlea [93]. Above the acoustics are phonemes, the smallest contrastive speech units in a language (e.g., /b/ versus /p/). Individual phonemes have variable duration and typically last around 50–150 ms [20]. Phonemes combine to form syllables, such as “ba” or “da”. Syllables are in turn combined into words, which provide lexical meaning and typically unfold over several hundred milliseconds. Although word durations vary substantially, most spoken words last between approximately 200–600 ms [7,8].

Words combine into phrases and sentences, introducing syntactic structure and compositional meaning, and typically unfold over timescales of hundreds of milliseconds to several seconds. At the highest levels are the broader discourse, which spans information across extended stretches of speech. These representations support the construction of coherent interpretations that can persist over many seconds, minutes, or even longer [94].

This representational hierarchy is accompanied by a broad anatomical hierarchy across the cortex [95–97]. Acoustic and phonetic features are represented most strongly in the auditory cortex and surrounding superior temporal regions [98]. Progressively more anterior and lateral temporal regions are sensitive to increasingly abstract lexical and semantic information [99]. Combinatorial processes that integrate words into phrases and sentences additionally recruit anterior temporal and inferior frontal regions [100], while discourse and narrative representations engage a widely distributed network extending beyond classical language-selective cortex [101].

Representational levels do not map onto discrete cortical areas in a simple one-to-one fashion. Although speech processing is supported by a broad anatomical hierarchy, the same neural populations, and even individual neurons, have been shown to encode multiple speech features [21,53,102]. Its spatial structure is therefore distributed and overlapping, while retaining broad hierarchical anatomical organisation.

Box 2: Speech-specific versus domain-general mechanism

The principles of persistent, parallel, and time-stamped encoding hold many computational advantages for the speech comprehension system, but may not be speech-specific.

Text is less transient than speech, but reading and listening share broader language processing demands, including the need to assemble higher-order structure from sequential elements. Parallel processing has been observed during reading: neural responses encode upcoming parafoveal words while the fixated word is still being processed [103,104], and information across the hierarchy is encoded simultaneously [105]. Whether reading also relies on similar persistent and time-stamped encoding mechanisms remains to be established.

Evidence from other sensory and cognitive domains provides clear analogues of all three principles. Persistent encoding has been well established in vision, finding that a briefly presented image remains decodable after it has disappeared, and can persist while subsequent images are presented [106,107]. Parallel encoding has also been observed during rapid visual presentation, where multiple successive images can be decoded concurrently from neural activity [108]. Similarly, during motor-sequence planning, neural populations simultaneously represent several component actions before their execution [109,110]. Time-stamped encoding has been identified across several species. In rats, hippocampal time cells fire at successive moments within an unfolding interval, collectively encoding elapsed time [75,111]. In songbirds, individual HVC neurons fire at precise moments within the song, forming an explicit population code for temporal position in the sequence [76]. Finding that shared principles support human speech processing is therefore a powerful extension of prior work, pointing to a set of computational mechanisms, preserved across species, which have been recycled to uphold a range of sensory and cognitive processes.

A related question concerns the relationship between persistent encoding and working memory. Traditional working-memory paradigms require information to be maintained across a sensorily silent delay. Speech comprehension instead requires previous information to remain available while being continuously bombarded with new input. The latter resembles “process memory” rather than traditional working memory [112], in which memory is intrinsic to ongoing processing: the same circuits process incoming information while retaining information about previous inputs. Persistent linguistic representations could therefore arise within the language-processing hierarchy without being transferred into a separate working-memory store. However, domain-general working-memory systems may additionally modulate these representations when information must be maintained over longer intervals, or manipulated according to artificial task goals.

Box 3: Dynamics of linguistic and statistical structure

Not all moments in spoken language place the same demands on, or provide the same amount of information to, the comprehension system. Some inputs are predictable from the preceding context, whereas others provide uncertainty about what is coming next; some inputs can be interpreted locally, whereas others depend on the integration of non-adjacent elements. These differences shape the neural dynamics of speech processing.

The processing of an input may begin earlier or later depending on the context in which it occurs. For example, recent studies have found that predictability, quantified as surprisal [14], affects the timing of processing across different levels of the linguistic hierarchy. At the phonetic level, predictable phonemes are decodable earlier than unpredictable ones [5]; at the lexical level, predictable words elicit earlier and smaller N400-like responses, whereas high-surprisal words produce later and larger responses [90]; at the syntactic level, a highly surprising word can delay syntactic structure building by approximately 150–190 ms [113].

Other aspects of the speech signal lead to information being processed for a shorter or longer period of time. For example, entropy describes uncertainty about which element will occur, and has been found to modulate the longevity of information encoding. Phonetic information is maintained longer while word identity remains uncertain, suggesting that lower-level detail is preserved until lexical entropy has been sufficiently reduced [5,6,114]. Syntactic structure also modulates longevity of encoding. For example, in the sentence “*Which book did the student say the teacher recommended?*”, “which book” cannot be assigned its role until “recommended” is heard. Neural activity remains sustained while this relationship is unresolved, followed by a P600 response when the two elements can be integrated. This response occurs later when the elements are separated by two clauses rather than one, suggesting that longer syntactic dependencies delay their resolution [115].

Together, these findings suggest that the timing of a given neural process is not fully determined by stimulus onset, or governed by a fixed processing schedule. Instead, it adapts to the informational and computational demands of each moment. Predictability influences when representations become available, whereas uncertainty and unresolved linguistic structure influence how long information remains available and when it can be integrated. Thus, linguistic context shapes not only what is represented and how strongly, but also when and for how long it is processed.

Glossary

Composition: Building higher-order representations from smaller linguistic elements.

Comprehension: Deriving the same meaning of an utterance as the speaker intended.

Hierarchical structure: The nested organisation of linguistic units into progressively larger structures.

Neural dimension: A coordinated pattern of activity across a set of neurons, electrodes, or sensors, along which information can vary or be encoded.

Neural trajectory: The path traced through a state space over time.

Parallel encoding: The simultaneous representation of multiple inputs or multiple levels of linguistic structure.

Persistent encoding: Neural representations that outlast the sensory events that generated them.

Postdictive processing: The use of later information to update the interpretation of earlier input.

Recoverable representations: Information that can be read out from neural activity and used for downstream computation, even when multiple representations are encoded simultaneously.

Representational format: The way information is encoded in neural activity, defined by which properties are preserved, discarded, or emphasised. For example, a speech sound may be represented in “acoustic format” or more abstract “phoneme category format”.

Rotational dynamics: The process of transferring neurally encoded information from one neural dimension to another over time.

Speech hierarchy: The organisation of speech into multiple levels, from sounds to words, phrases, and meaning.

State space: An abstract space in which each coordinate corresponds to a pattern of activity of a neural population.

Time-code: A representation that encodes both the content of an element and when it occurred.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used GPT 5.5 to improve clarity of explanation, and reduce the word count. The author edited the output as needed and takes full responsibility for the content of the published article.

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References

1. Picton, T.W. *et al.* (1974) Human auditory evoked potentials. I. Evaluation of components. *Electroencephalogr. Clin. Neurophysiol.* 36, 179–190
2. Sams, M. *et al.* (1993) The human auditory sensory memory trace persists about 10 sec: Neuromagnetic evidence. *J. Cogn. Neurosci.* 5, 363–370
3. Klatt, D.H. (1976) Linguistic uses of segmental duration in English: acoustic and perceptual evidence. *J. Acoust. Soc. Am.* 59, 1208–1221
4. Khalighinejad, B. *et al.* (2017) Dynamic encoding of acoustic features in neural responses to continuous speech. *J. Neurosci.* 37, 2176–2185
5. Gwilliams, L. *et al.* (2022) Neural dynamics of phoneme sequences reveal position-invariant code for content and order. *Nat. Commun.* 13, 6606
6. Gwilliams, L. *et al.* (2017) In spoken word recognition, the future predicts the past. *J. Neurosci.* 38, 7585–7599
7. Tauroza, S. and Allison, D. (1990) Speech Rates in British English. *Appl. Linguist.* 11, 90–105
8. Jusezyk, P.W. and Luce, P.A. (2002) Speech perception and spoken word recognition: past and present. *Ear Hear.* 23, 2–40
9. Gwilliams, L. *et al.* (2025) Hierarchical dynamic coding coordinates speech comprehension in the human brain. *Proc. Natl. Acad. Sci. U. S. A.* 122, e2422097122
10. Heilbron, M. *et al.* (2022) A hierarchy of linguistic predictions during natural language comprehension. *Proc. Natl. Acad. Sci. U. S. A.* 119, e2201968119
11. Karunathilake, I. *et al.* (2025) Neural dynamics of the processing of speech features: Evidence for a progression of features from acoustic to sentential processing. *J. Neurosci.* 45
12. Gwilliams, L. *et al.* (2023) Top-down information shapes lexical processing when listening to continuous speech. *Language, Cognition and Neuroscience* DOI: [10.1080/23273798.2023.2171072](https://doi.org/10.1080/23273798.2023.2171072)
13. Chan, A.M. *et al.* (2011) Decoding word and category-specific spatiotemporal representations from MEG and EEG. *Neuroimage* 54, 3028–3039
14. Gwilliams, L. and Davis, M.H. (2022) Extracting language content from speech sounds: The information theoretic approach. In *Speech Perception*, pp. 113–139, Springer International Publishing
15. d’Ascoli, S. *et al.* (2025) Towards decoding individual words from non-invasive brain recordings. *Nat. Commun.* 16, 10521
16. Goldstein, A. *et al.* (2022) Shared computational principles for language processing in humans and deep language models. *Nat. Neurosci.* 25, 369–380
17. Gaston, P. and Marantz, A. (2018) The time course of contextual cohort effects in auditory processing of category-ambiguous words: MEG evidence for a single “clash” as noun or verb. *Lang. Cogn. Neurosci.* 33, 402–423
18. Ettinger, A. *et al.* (2014) The role of morphology in phoneme prediction: evidence from MEG. *Brain Lang.* 129, 14–23
19. Matar, S., & Marantz, A. (2025). Neural bases of proactive and predictive processing of meaningful subword units in speech comprehension. *The Journal of Neuroscience*, 45(7), e0781242024.
20. Norman-Haignere, S.V. *et al.* (2025) Temporal integration in human auditory cortex is predominantly yoked to absolute time. *Nat. Neurosci.* 28, 2356–2365
21. Keshishian, M. *et al.* (2023) Joint, distributed and hierarchically organized encoding of

- linguistic features in the human auditory cortex. *Nat. Hum. Behav.* 7, 740–753
22. Rodd, J. (2017) Lexical ambiguity. In *Oxford Handbook of Psycholinguistics* (Gaskell, M. G. and Rueschemeyer, S.-A., eds), Oxford University Press
 23. MacDonald, M.C. *et al.* (2015) Syntactic Ambiguity Resolution as Lexical Ambiguity Resolution. In *Perspectives on Sentence Processing* (1st Edition.), pp. 123–153, Psychology Press
 24. de Heer Kloots, M. *et al.* (2026) Distinct yet neighboring neural populations encode past, future, and surrounding speech context in the human temporal lobe. *bioRxiv*, DOI: 10.64898/2026.05.13.724774
 25. Pickering, M.J. and Gambi, C. (2018) Predicting while comprehending language: A theory and review. *Psychol. Bull.* 144, 1002–1044
 26. Ryskin, R. and Nieuwland, M.S. (2023) Prediction during language comprehension: what is next? *Trends Cogn. Sci.* 27, 1032–1052
 27. Lau, E.F. *et al.* (2008) A cortical network for semantics: (de)constructing the N400. *Nat. Rev. Neurosci.* 9, 920–933
 28. Donhauser, P.W. and Baillet, S. (2020) Two distinct neural timescales for predictive speech processing. *Neuron* 105, 385–393.e9
 29. Ganong, W.F. (1980) Phonetic categorization in auditory word perception. *J. Exp. Psychol. Hum. Percept. Perform.* 6, 110–125
 30. Connine, C. *et al.* (1991) Effects of subsequent sentence context in auditory word recognition: Temporal and linguistic constraint. *Journal of Memory and Language* 30, 234–250
 31. McMurray, B. *et al.* (2009) Within-category VOT affects recovery from “lexical” garden paths: Evidence against phoneme-level inhibition. *J. Mem. Lang.* 60, 65–91
 32. Turner, W., Hogendoorn, H., & Gwilliams, L. (2026). On the speed of conscious perception: how soon is now?. *The Behavioral and brain sciences*, 49, e195.
<https://doi.org/10.1017/S0140525X25103051>
 33. McClelland, J.L. and Elman, J.L. (1986) The TRACE model of speech perception. *Cogn. Psychol.* 18, 1–86
 34. Bicknell, K. *et al.* (2025) Maintenance of subcategorical information during speech perception: revisiting misunderstood limitations. *J. Mem. Lang.* 140, 104565
 35. Gwilliams, L., Poeppel, D., Marantz, A., & Linzen, T. (2018, January). Phonological (un)certainly weights lexical activation. In *Proceedings of the 8th workshop on cognitive modeling and computational linguistics (cmcl 2018)* (pp. 29-34). DOI: <https://doi.org/10.48550/arXiv.1711.06729>
 36. Christiansen, M.H. and Chater, N. (2016) The Now-or-Never bottleneck: A fundamental constraint on language. *Behav. Brain Sci.* 39, e62
 37. Honey, C.J. *et al.* (2016) How long is now? The multiple timescales of language processing. *Behav. Brain Sci.* 39, e77
 38. Mollo, G. *et al.* (2018) Context-dependent lexical ambiguity resolution: MEG evidence for the time-course of activity in left inferior frontal gyrus and posterior middle temporal gyrus. *Brain and language* 177, 23–36
 39. MacGregor, L.J. *et al.* (2020) The neural time course of semantic ambiguity resolution in speech comprehension. *Journal of cognitive neuroscience* 32, 403–425
 40. Osterhout, L. and Holcomb, P.J. (1992) Event-related brain potentials elicited by syntactic anomaly. *J. Mem. Lang.* 31, 785–806
 41. Friederici, A.D. *et al.* (1996) Temporal structure of syntactic parsing: Early and late event-related brain potential effects. *J. Exp. Psychol. Learn. Mem. Cogn.* 22, 1219–1248
 42. Mason, R.A. *et al.* (2003) Ambiguity in the brain: what brain imaging reveals about the processing of syntactically ambiguous sentences. *J. Exp. Psychol. Learn. Mem. Cogn.* 29, 1319–1338

43. den Ouden, D.-B. *et al.* (2016) Neural correlates of early-closure garden-path processing: Effects of prosody and plausibility. *Q. J. Exp. Psychol. (Hove)* 69, 926–949
44. Xu, Y. and Huang, X. (2025) Neural mechanisms of dynamic syntactic and semantic processing in Chinese garden-path sentence comprehension: An ERP study. *J. Neurolinguistics* 73, 101233
45. Christianson, K. *et al.* (2001) Thematic roles assigned along the garden path linger. *Cogn. Psychol.* 42, 368–407
46. Ferreira, F. and Patson, N.D. (2007) The “good enough” approach to language comprehension: The “good enough” approach. *Lang. Linguist. Compass* 1, 71–83
47. McClelland, J.L. (2016) The case for interactionism in language processing. *Attention and performance XII* at <https://api.taylorfrancis.com/content/chapters/edit/download?identifierName=doi&identifierValue=10.4324/9781315630427-2&type=chapterpdf>
48. Keshishian, M. *et al.* (2026) Parallel hierarchical encoding of linguistic representations in the human auditory cortex and recurrent automatic speech recognition systems. *Nat. Mach. Intell.* 8, 257–269
49. Shannon, C.E. (1951) Prediction and entropy of printed English. *Bell Syst. Tech. J.* 30, 50–64
50. Jelinek, F. (1990) Self-organized language modeling for speech recognition. In *Readings in Speech Recognition*, pp. 450–506, Elsevier
51. Jurafsky, D. and Martin, J.H. (2000) Speech and language processing: An introduction to natural language processing, computational linguistics, and speech recognition. *aclanthology.org* at <https://aclanthology.org/anthology-files/pdf/J/J00/J00-4006.pdf>
52. Bai, F. *et al.* (2022) Neural dynamics differentially encode phrases and sentences during spoken language comprehension. *PLoS Biol.* 20, e3001713
53. Brodbeck, C. *et al.* (2022) Parallel processing in speech perception with local and global representations of linguistic context. *Elife* 11
54. Desbordes, T. *et al.* (2024) Tracking the neural codes for words and phrases during semantic composition, working-memory storage, and retrieval. *Cell Rep.* 43, 113847
55. Martin, A.E. and Doumas, L.A.A. (2017) A mechanism for the cortical computation of hierarchical linguistic structure. *PLoS Biol.* 15, e2000663
56. Ding, N. (2020) A structure-based memory maintenance model for neural tracking of linguistic structures, *arXiv*. DOI: <https://doi.org/10.48550/arXiv.2002.11870>
57. Lyu, B. *et al.* (2024) Finding structure during incremental speech comprehension. *Elife* 12, RP89311
58. Libby, A. and Buschman, T.J. (2021) Rotational dynamics reduce interference between sensory and memory representations. *Nat. Neurosci.* 24, 715–726
59. de Heer, W.A. *et al.* (2017) The hierarchical cortical organization of human speech processing. *J. Neurosci.* 37, 6539–6557
60. Friederici, A.D. (2012) The cortical language circuit: from auditory perception to sentence comprehension. *Trends Cogn. Sci.* 16, 262–268
61. Eisner, F. and McQueen, J.M. (2018) Speech perception. *Stevens' handbook of experimental psychology and cognitive neuroscience* 3, 1–46
62. Sohoglu, E. *et al.* (2012) Predictive top-down integration of prior knowledge during speech perception. *J. Neurosci.* 32, 8443–8453
63. Davis, M.H. *et al.* (2011) Does semantic context benefit speech understanding through “top-down” processes? Evidence from time-resolved sparse fMRI. *Journal of cognitive neuroscience* 23, 3914–3932
64. Boudewyn, M.A. *et al.* (2019) Adaptation to Animacy Violations during Listening Comprehension. *Cogn. Affect. Behav. Neurosci.* 19, 1247–1258
65. Marshuetz, C. (2005) Order information in working memory: an integrative review of

- evidence from brain and behavior. *Psychol. Bull.* 131, 323–339
66. Hurlstone, M.J. *et al.* (2014) Memory for serial order across domains: An overview of the literature and directions for future research. *Psychol. Bull.* 140, 339–373
 67. Farrell, S. (2012) Temporal clustering and sequencing in short-term memory and episodic memory. *Psychol. Rev.* 119, 223–271
 68. Ferbinteanu, J. *et al.* (2006) Episodic memory--from brain to mind. *Hippocampus* 16, 691–703
 69. Foster, D.J. and Knierim, J.J. (2012) Sequence learning and the role of the hippocampus in rodent navigation. *Curr. Opin. Neurobiol.* 22, 294–300
 70. Singer, P. *et al.* (2014) Detecting memory and structure in human navigation patterns using Markov chain models of varying order. *PLoS One* 9, e102070
 71. Tanji, J. (2001) Sequential organization of multiple movements: involvement of cortical motor areas. *Annu. Rev. Neurosci.* 24, 631–651
 72. Doyon, J. *et al.* (2018) Current issues related to motor sequence learning in humans. *Current opinion in behavioral sciences* 20, 89–97
 73. MacDonald, C.J. *et al.* (2011) Hippocampal “time cells” bridge the gap in memory for discontinuous events. *Neuron* 71, 737–749
 74. Eichenbaum, H. (2014) Time cells in the hippocampus: a new dimension for mapping memories. *Nat. Rev. Neurosci.* 15, 732–744
 75. MacDonald, C.J. *et al.* (2013) Distinct hippocampal time cell sequences represent odor memories in immobilized rats. *J. Neurosci.* 33, 14607–14616
 76. Hahnloser, R.H.R. *et al.* (2002) An ultra-sparse code underlies the generation of neural sequences in a songbird. *Nature* 419, 65–70
 77. Long, M.A. *et al.* (2010) Support for a synaptic chain model of neuronal sequence generation. *Nature* 468, 394–399
 78. Zhang, Y. *et al.* (2026) Human cortical dynamics of auditory word form encoding. *Neuron* 114, 167–180.e6
 79. Ding, N. (2025) Sequence chunking through neural encoding of ordinal positions. *Trends Cogn. Sci.* DOI: [10.1016/j.tics.2025.01.014](https://doi.org/10.1016/j.tics.2025.01.014)
 80. Solway, A. *et al.* (2012) Positional and temporal clustering in serial order memory. *Mem. Cognit.* 40, 177–190
 81. Henson, R.N.A. (1999) Coding position in short-term memory. *Int. J. Psychol.* 34, 403–409
 82. Gomez, P. *et al.* (2008) The overlap model: a model of letter position coding. *Psychol. Rev.* 115, 577–600
 83. Vaswani, A. *et al.* (2017) Attention Is All You Need, *arXiv*: DOI: <https://doi.org/10.48550/arXiv.1706.03762>
 84. Su, J. *et al.* (2024) RoFormer: Enhanced transformer with Rotary Position Embedding. *Neurocomputing* 568, 127063
 85. Peng, B. *et al.* (2023) YaRN: Efficient Context Window Extension of Large Language Models, *arXiv*. DOI: <https://doi.org/10.48550/arXiv.2309.00071>
 86. Liu, O.D. *et al.* (2026) Brain-like dynamics in speech representations can emerge through self-supervised learning, *bioRxiv*. DOI: <https://doi.org/10.64898/2026.01.16.700011>
 87. Paulk, A.C. *et al.* (2022) Large-scale neural recordings with single neuron resolution using Neuropixels probes in human cortex. *Nat. Neurosci.* 25, 252–263
 88. Chung, J.E. *et al.* (2022) High-density single-unit human cortical recordings using the Neuropixels probe. *Neuron* 110, 2409–2421.e3
 89. Jun, J.J. *et al.* (2017) Fully integrated silicon probes for high-density recording of neural activity. *Nature* 551, 232–236
 90. Dou, J. *et al.* (2025) Dynamic modeling of EEG responses to natural speech reveals earlier processing of predictable words. *PLoS Comput. Biol.* 21, e1013006
 91. Ergin, I. *et al.* (2026) Measuring naturalistic speech comprehension in real time. *Behav.*

- Res. Methods* 58, 93
92. Gwilliams, L. *et al.* (2025) Computational Architecture of Speech Comprehension in the Human Brain. 11
 93. Robles, L. and Ruggero, M.A. (2001) Mechanics of the mammalian cochlea. *Physiol. Rev.* 81, 1305–1352
 94. Yeshurun, Y. *et al.* (2021) The default mode network: where the idiosyncratic self meets the shared social world. *Nat. Rev. Neurosci.* 22, 181–192
 95. Hasson, U. *et al.* (2008) A hierarchy of temporal receptive windows in human cortex. *J. Neurosci.* 28, 2539–2550
 96. Honey, C.J. *et al.* (2012) Slow cortical dynamics and the accumulation of information over long timescales. *Neuron* 76, 423–434
 97. Lerner, Y. *et al.* (2011) Topographic mapping of a hierarchy of temporal receptive windows using a narrated story. *J. Neurosci.* 31, 2906–2915
 98. Bhaya-Grossman, I. and Chang, E.F. (2022) Speech computations of the human superior temporal gyrus. *Annu. Rev. Psychol.* 73, 79–102
 99. Hickok, G. (2022) The dual stream model of speech and language processing. *Handb. Clin. Neurol.* 185, 57–69
 100. Pyllkkänen, L. (2019) The neural basis of combinatory syntax and semantics. *Science* 366, 62–66
 101. Fedorenko, E. *et al.* (2024) The language network as a natural kind within the broader landscape of the human brain. *Nat. Rev. Neurosci.* DOI: [10.1038/s41583-024-00802-4](https://doi.org/10.1038/s41583-024-00802-4)
 102. Leonard, M.K. *et al.* (2023) Large-scale single-neuron speech sound encoding across the depth of human cortex. *Nature* 626, 593–602
 103. Pan, Y. *et al.* (2021) Neural evidence for lexical parafoveal processing. *Nat. Commun.* 12, 5234
 104. Pan, Y. *et al.* (2024) Early parafoveal semantic integration in natural reading. *Elife* 12
 105. Raghavan, V.S. and Parra, L.C. (2025) Neural encoding of linguistic features during natural sentence reading. *iScience* 28, 112798
 106. Carlson, T. *et al.* (2013) Representational dynamics of object vision: the first 1000 ms. *J. Vis.* 13, 1–1
 107. Grootswagers, T. *et al.* (2019) The representational dynamics of visual objects in rapid serial visual processing streams. *Neuroimage* 188, 668–679
 108. King, J.-R. and Wyart, V. (2021) The Human Brain Encodes a Chronicle of Visual Events at Each Instant of Time Through the Multiplexing of Traveling Waves. *J. Neurosci.* 41, 7224–7233
 109. Averbach, B.B. *et al.* (2002) Parallel processing of serial movements in prefrontal cortex. *Proc. Natl. Acad. Sci. U. S. A.* 99, 13172–13177
 110. Kornysheva, K. *et al.* (2019) Neural competitive queuing of ordinal structure underlies skilled sequential action. *Neuron* 101, 1166–1180.e3
 111. Kraus, B.J. *et al.* (2013) Hippocampal “time cells”: time versus path integration. *Neuron* 78, 1090–1101
 112. Hasson, U. *et al.* (2015) Hierarchical process memory: memory as an integral component of information processing. *Trends Cogn. Sci.* 19, 304–313
 113. Slaats, S. *et al.* (2024) Lexical surprisal shapes the time course of syntactic structure building. *Neurobiol. Lang. (Camb.)* 5, 942–980
 114. Kries, J. *et al.* (2026) The spatio-temporal dynamics of phoneme encoding in aging and aphasia. *J. Neurosci.* 46, e1001252025
 115. Phillips, C. *et al.* (2005) ERP effects of the processing of syntactic long-distance dependencies. *Cognitive Brain Research* 22, 407–428